Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science focused on building systems that can perform tasks that normally require human intelligence. These tasks include understanding language, recognizing images, solving problems, and making decisions.

Machine Learning is a subset of AI that allows computers to learn patterns from data instead of being explicitly programmed. Popular types of machine learning include supervised learning, unsupervised learning, and reinforcement learning.

Deep Learning is a specialized area of machine learning that uses neural networks with many layers. It is commonly used in applications such as speech recognition, computer vision, and natural language processing.

Large Language Models (LLMs) are deep learning models trained on massive text datasets. They can generate human-like text, answer questions, summarize documents, and assist with many knowledge-based tasks.

Retrieval Augmented Generation (RAG) is a technique that improves LLM responses by retrieving relevant information from external documents before generating an answer. This allows the system to provide more accurate and up-to-date responses.